SMALL AND BIG PUSHFORWARDS AGREE FOR PROPER MORPHISMS, FOR TORSION COEFFICIENTS AND FOR Z/NZ COEFFICIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full for all four parts: the formal unwinding in (1), the pullback computation identifying the base change map in (2), the same argument with modulo n coefficients in (3), and the exactness reductions for finite morphisms in (4).

Editorial note. The label lemma-compare-higher-direct-image-proper names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; the seven cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The comparison morphisms pi_S and the functors f_small,*, f_big,*. Same chapter, the morphism pi_S : (Sch/S)_etale -> S_etale and the small and big pushforwards of a morphism of schemes f : T -> S. Supplies both sides of every comparison asserted in the four parts.
- Base change of the comparison map along an object of (Sch/S)_etale. Cohomology on Sites, the remark providing the canonical base change map pi_S^{-1}Rf_{small,*}K -> Rf_{big,*}pi_T^{-1}K. Supplies the map whose being an isomorphism is proved in parts (2) and (3).
- Proper base change for torsion coefficients and modulo n coefficients. Same chapter, the proper base change theorem and its Z/nZ variant. Supplies the isomorphism of the pulled-back base change map in parts (2) and (3).
- Exactness of the pushforward along a finite morphism. Same chapter, the vanishing of the higher direct images of a finite morphism. Supplies exactness of f_{small,*} and, via the comparison lemma, that of f_{big,*} in part (4).

Source reference. Original lemma 075A: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21196–21196. Prerequisite pointer invoked at proof line(s) 21393; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 09XN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4960–4960. Prerequisite pointer invoked at proof line(s) 21354; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09V5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18708–18708. Prerequisite pointer invoked at proof line(s) 21395; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A3U: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18282–18282. Prerequisite pointer invoked at proof line(s) 21362; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0F0C: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18887–18887. Prerequisite pointer invoked at proof line(s) 21400; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 03QP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8245–8245. Prerequisite pointer invoked at proof line(s) 21406; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 07A7: sources/stacks-chow/tags.txt, source lines 9297–9297. Prerequisite pointer invoked at proof line(s) 21374; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : T \to S$ be a proper morphism of schemes. Then we have*

1. $\pi_S^{-1} \circ f_{small,*} = f_{big,*} \circ \pi_T^{-1}$ *as functors* $Sh(T_{étale}) \to Sh((Sch/S)_{étale})$,
2. $\pi_S^{-1} Rf_{small,*} K = Rf_{big,*} \pi_T^{-1} K$ *for* K *in* $D^+(T_{étale})$ *whose cohomology sheaves are torsion,*
3. $\pi_S^{-1} Rf_{small,*} K = Rf_{big,*} \pi_T^{-1} K$ *for* K *in* $D(T_{étale}, \mathbf{Z}/n\mathbf{Z})$, *and*
4. $\pi_S^{-1} Rf_{small,*} K = Rf_{big,*} \pi_T^{-1} K$ *for all* K *in* $D(T_{étale})$ *if* f *is finite.*

Proof. Proof of (1). Let $\mathcal{F}$ be a sheaf on $T_{étale}$. Let $g : S' \to S$ be an object of $(Sch/S)_{étale}$. Consider the fibre product

$$\begin{array}{ccc} T' & \xrightarrow{f'} & S' \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ T & \xrightarrow{f} & S \end{array}$$

Then we have

$$(f_{big,*}\pi_T^{-1}\mathcal{F})(S') = (\pi_T^{-1}\mathcal{F})(T') = ((g'_{small})^{-1}\mathcal{F})(T') = (f'_{small,*}(g'_{small})^{-1}\mathcal{F})(S')$$

the second equality by Lemma 09XN. On the other hand

$$(\pi_S^{-1} f_{small,*}\mathcal{F})(S') = (g_{small}^{-1} f_{small,*}\mathcal{F})(S')$$

again by Lemma 09XN. Hence by proper base change for sheaves of sets (Lemma 0A3U) we conclude the two sets are canonically isomorphic. The isomorphism is

compatible with restriction mappings and defines an isomorphism $\pi_S^{-1} f_{small,*}\mathcal{F} = f_{big,*}\pi_T^{-1}\mathcal{F}$. Thus an isomorphism of functors $\pi_S^{-1} \circ f_{small,*} = f_{big,*} \circ \pi_T^{-1}$.

Proof of (2). There is a canonical base change map $\pi_S^{-1} Rf_{small,*}K \to Rf_{big,*}\pi_T^{-1}K$ for any K in $D(T_{étale})$, see Cohomology on Sites, Remark 07A7. To prove it is an isomorphism, it suffices to prove the pull back of the base change map by $i_g : Sh(S'_{étale}) \to Sh((Sch/S)_{étale})$ is an isomorphism for any object $g : S' \to S$ of $(Sch/S)_{étale}$. Let T', g', f' be as in the previous paragraph. The pullback of the base change map is

$$\begin{aligned} g_{small}^{-1} Rf_{small,*}K &= i_g^{-1}\pi_S^{-1} Rf_{small,*}K \\ &\to i_g^{-1} Rf_{big,*}\pi_T^{-1}K \\ &= Rf'_{small,*}(i_{g'}^{-1}\pi_T^{-1}K) \\ &= Rf'_{small,*}((g'_{small})^{-1}K) \end{aligned}$$

where we have used $\pi_S \circ i_g = g_{small}$, $\pi_T \circ i_{g'} = g'_{small}$, and Lemma 075A. This map is an isomorphism by the proper base change theorem (Lemma 09V5) provided K is bounded below and the cohomology sheaves of K are torsion.

The proof of part (3) is the same as the proof of part (2), except we use Lemma 0F0C instead of Lemma 09V5.

Proof of (4). If f is finite, then the functors $f_{small,*}$ and $f_{big,*}$ are exact. This follows from Proposition 03QP for f_{small}. Since any base change f' of f is finite too, we conclude from Lemma 075A part (3) that $f_{big,*}$ is exact too (as the higher derived functors are zero). Thus this case follows from part (1). □